UNIT 3 · 7 hours

# Correlation Between Time and Frequency Responses



Frequency Domain Analysis

**Control Systems (22EEG53)**
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# Learning objectives

- Relate Mp and Mr through ζ
- Relate bandwidth to rise and settling time
- Relate phase margin to damping ratio
- Estimate time-domain performance from frequency plots

**Syllabus context**

Frequency domain analysis: frequency domain specifications peak resonance, resonant frequency, bandwidth and cut-off rate; correlation between time and frequency responses for second-order systems; Bode plot, gain margin and phase margin.

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# Why correlate?

- Both responses are determined by the same parameters ωn and ζ.
- Frequency tests are easy; time-domain specs are what the user cares about.
- For a second-order (or dominant second-order) system exact formulas relate them.
- Design may be done in either domain.

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# Peak overshoot Mp versus resonant peak Mr

| ζ | Mr | Mp (%) |
|---|---|---|
| **0.2** | 2.55 | 52.7 |
| **0.3** | 1.75 | 37.2 |
| **0.4** | 1.36 | 25.4 |
| **0.5** | 1.15 | 16.3 |
| **0.6** | 1.04 | 9.5 |
| **0.707** | 1.00 | 4.3 |

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# Other correlations

### Bandwidth and speed

$\omega_b \propto \omega_n$, so $t_r \propto 1/\omega_b$: larger bandwidth gives smaller rise time.

### ωr and tp

$t_p = \pi/\omega_d$, and $\omega_r \approx \omega_n\sqrt{(1 - 2\zeta^2)}$: larger ωr gives smaller peak time.

### Settling

$t_s = 4/(\zeta\omega_n)$: a large $\omega_n$ (large bandwidth) or damping gives a small ts.

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# Phase margin and damping ratio

| ζ | Phase margin PM (°) | Rule of thumb PM ≈ 100 ζ |
|---|---|---|
| **0.2** | 22.6 | 20 |
| **0.4** | 43.1 | 40 |
| **0.6** | 59.2 | 60 |
| **0.707** | 65.5 | 70 |

**Exact:** $PM = \tan^{-1}\{ 2\zeta / \sqrt{[\sqrt{(1 + 4\zeta^4)} - 2\zeta^2]} \}$

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# Numerical: use of correlation

**Problem** A unity-feedback system must have Mp ≤ 16 %. Estimate the required ζ, Mr limit and phase margin.

- Mp = 16 %: $\zeta \approx 0.5$
- $M_r = 1/(2\zeta\sqrt{(1-\zeta^2)}) = 1/(2 \times 0.5 \times 0.866) = 1.155$ (1.25 dB)
- $PM = \tan^{-1}[1/\sqrt{(\sqrt{1.25} - 0.5)}] = \tan^{-1}(1.272)$

**Answer: $\zeta \approx 0.5$, $M_r \approx 1.15$, PM ≈ 52°**

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# Key takeaways

- Mr and Mp both depend only on ζ
- tr ∝ 1/ωb; larger bandwidth = faster response
- PM ≈ 100 ζ for ζ < 0.6
- Use frequency plots to predict the transient behaviour

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# Practice questions

**Q1.** Find Mr for Mp = 25 %.

**Q2.** Why does a large bandwidth reduce rise time?

**Q3.** PM = 45°: estimate ζ and Mp.

**Q4.** Tabulate ζ, Mr, Mp for ζ = 0.2, 0.5.